

Team Worksheet — Lab 11

Continuous Gage R&R — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-11-gage-rr-continuous.xlsx`.

3. What we produced

Deliverable: A completed crossed Gage R&R study with %study variation and ndc for your capstone metric.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the components of measurement variation in full: observed variation = actual process variation + measurement system variation.

Step 2. Recap the five MSA properties: accuracy (bias), linearity, stability, repeatability (equipment) and reproducibility (operator).

Step 3. Distinguish repeatability from reproducibility precisely — same operator repeating, versus different operators on the same part.

Step 4. Design a crossed Gage R&R for your capstone metric: 10 parts spanning the full process range, 3 operators, 3 repeat measurements each.

Step 5. Select parts that span the real range of production. Parts clustered near the mean will flatter the gauge and hide its weakness.

Step 6. Randomise the measurement order and blind the operators to previous readings, or you will measure memory rather than the gauge.

Step 7. Run the study and record all 90 measurements in the standard crossed layout.

Step 8. Calculate %study variation for repeatability, reproducibility and total Gage R&R. Apply the acceptance rule: under 10% acceptable, 10-30% marginal, over 30% unacceptable.

Step 9. Calculate the number of distinct categories (ndc). ndc must be 5 or more for the gauge to distinguish parts usefully.

Step 10. If the gauge fails, decide the fix: recalibrate, retrain operators, tighten the operational definition or replace the instrument. Then re-run.

7. Self-check

Your study uses parts spanning the full process range with randomised blinded order, and you have computed %study variation and ndc with a stated accept/reject decision.

Complete? ■ Yes ■ Not yet

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